

AI FOR DESIGN

Introduction & Overview

<https://www.youtube.com/watch?v=98DcoXwGX6I>

WHAT IS DESIGN?

DESIGN

Design is a discipline of study and practice focused on the interaction between a person — a ‘user’ — and the man-made environment, taking into account aesthetic, functional, contextual, cultural and societal considerations.

- Artist
- Engineer
- Designer



DESIGN

Design is a discipline of study and practice focused on the interaction between a person — a ‘user’ — and the man-made environment, taking into account aesthetic, functional, contextual, cultural and societal considerations.

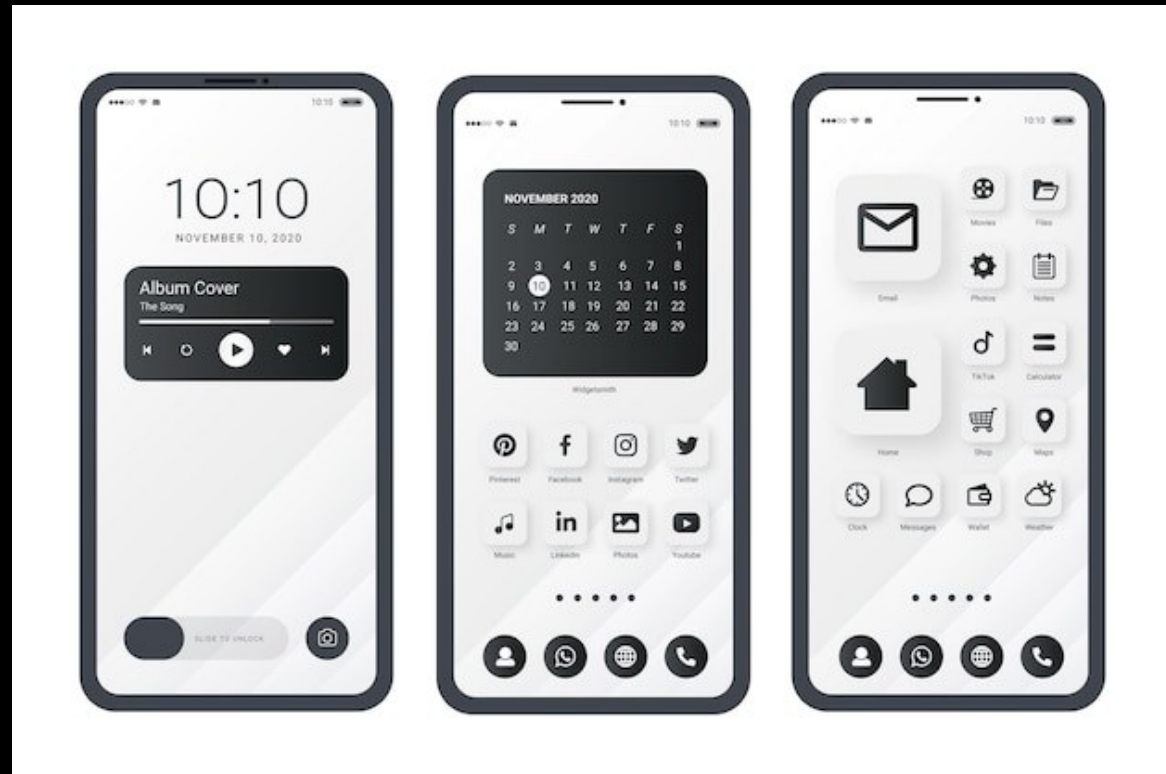
- Artist
- Engineer
- Designer



DESIGN

Design is a discipline of study and practice focused on the interaction between a person — a ‘user’— and the man-made environment, taking into account aesthetic, functional, contextual, cultural and societal considerations.

- Artist
- Engineer
- Designer



KEY PRINCIPLES OF DESIGN

- Form and function.

KEY PRINCIPLES OF DESIGN

- Form and function.
- Aesthetics and usability.

KEY PRINCIPLES OF DESIGN

- Form and function.
- Aesthetics and usability.
- Innovation and creativity.

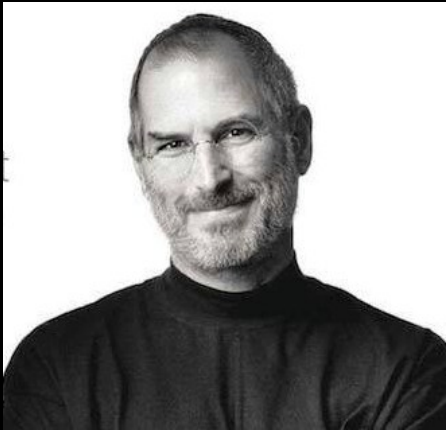
KEY PRINCIPLES OF DESIGN

- Form and function.
- Aesthetics and usability.
- Innovation and creativity.
- Sustainable design.

DIETER RAMS' 10 PRINCIPLES OF GOOD DESIGN

1. Good design is innovative
2. Good design makes a product useful
3. Good design is aesthetic
4. Good design makes a product understandable
5. Good design is unobtrusive
6. Good design is honest
7. Good design is long-lasting
8. Good design is thorough down to the last detail
9. Good design is environmentally friendly
10. Good design is as little design as possible

WHAT IS DESIGN?



Steve Jobs

"Design is not just what it looks like and feels like. Design is how it works."

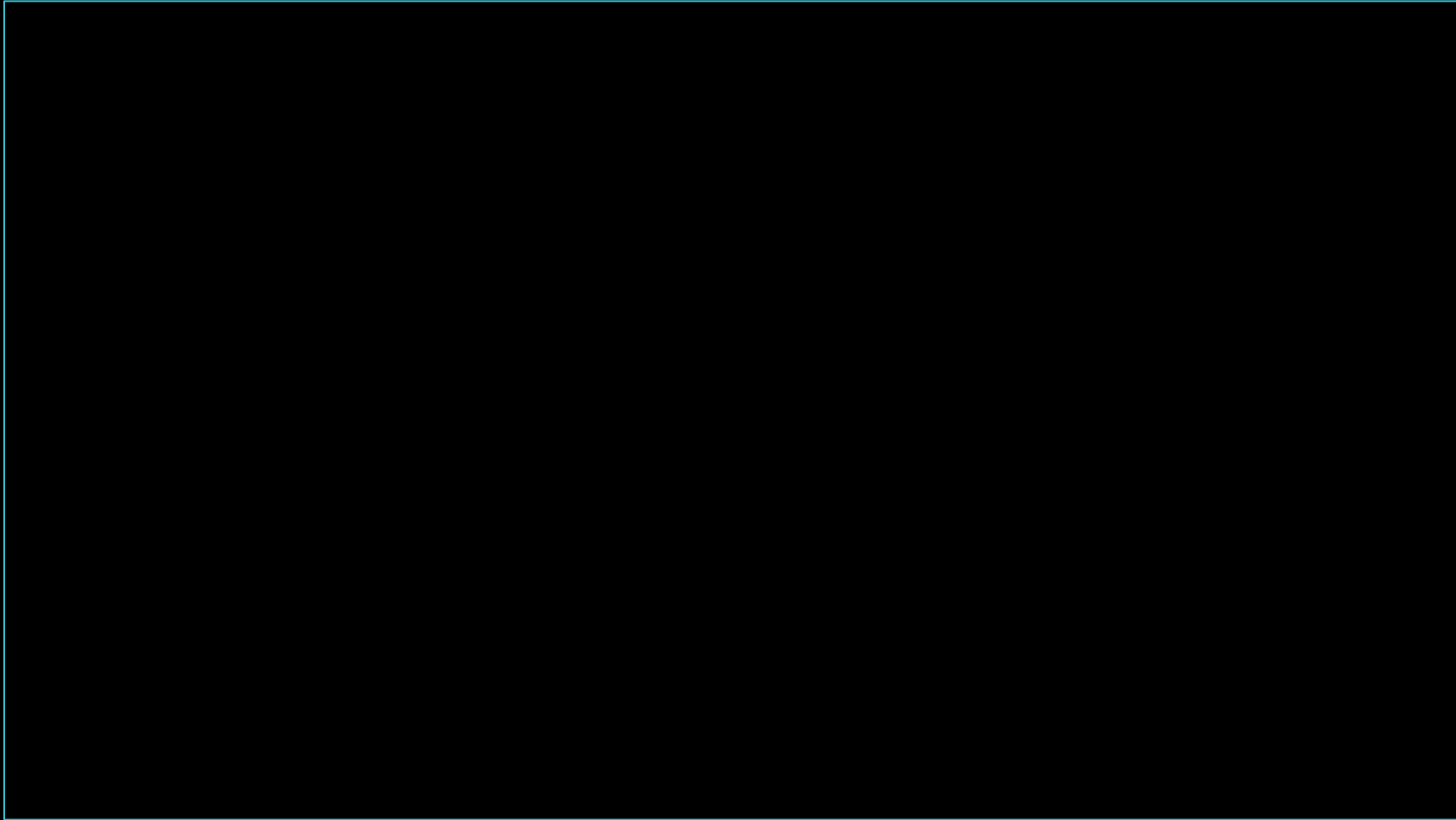


Tim Brown (IDEO):

"Design thinking is a human-centered approach to innovation that draws from the designer's toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success."

WHAT IS AI?

AI IS THE NEW HOT(EST) THING



Google I/O 2023 keynote in 24 seconds by @eddiegm

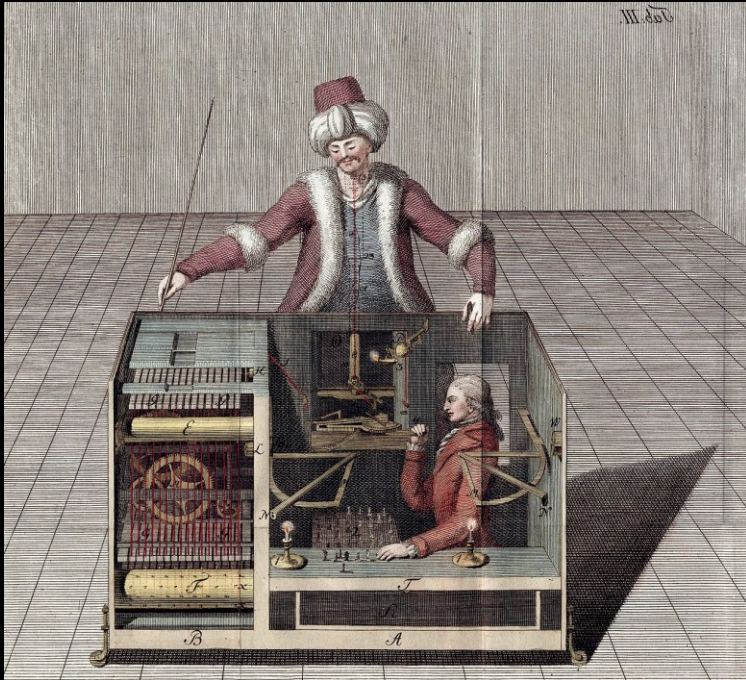
ARTIFICIAL INTELLIGENCE(AI)

- What is Artificial Intelligence?
- What is intelligence?



CHESS MACHINE

MECHANICAL TURK (1770)



DEEP BLUE VS GARRY KASPAROV (1977)



TESLER'S THEOREM

“Intelligence is whatever machines haven't done yet.”

-Larry Tesler



HUMAN INTELLIGENCE IS A LOT OF THINGS

- Reading
- Calculation
- Communication
- Reasoning
- Driving
- Operating
- Drawing
- Writing
- Creating
- Strategizing

_____ requires intelligence.

AI IS ALSO A LOT OF THINGS

Based on the Total Turing Test, an agent must have:

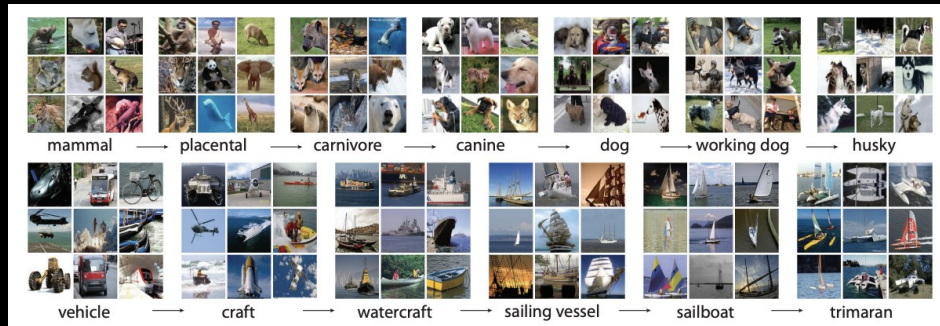
- Natural Language Processing
- Computer Vision
- Robotics
- Data Analysis
- Knowledge Representation
- Automated Reasoning
- ...



Deep Learning
(neural networks)

DEEP LEARNING/NEURAL NETWORK

ALEXNET(2012)



Top 5 Error Rate dropped from 25% to 16%.

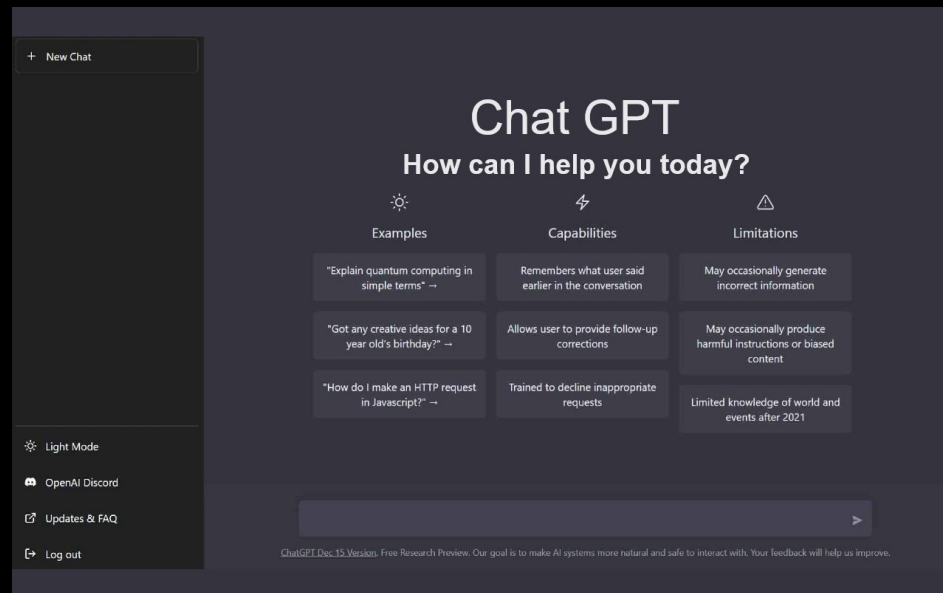
ALPHAGO(2016)



First time beating a 9-dan Go player with no handicap.

DEEP LEARNING/NEURAL NETWORK

CHAT-GPT (2022)



STABLE DIFFUSION (2022)



DEEP LEARNING/NEURAL NETWORK

GPT-40 (2024)



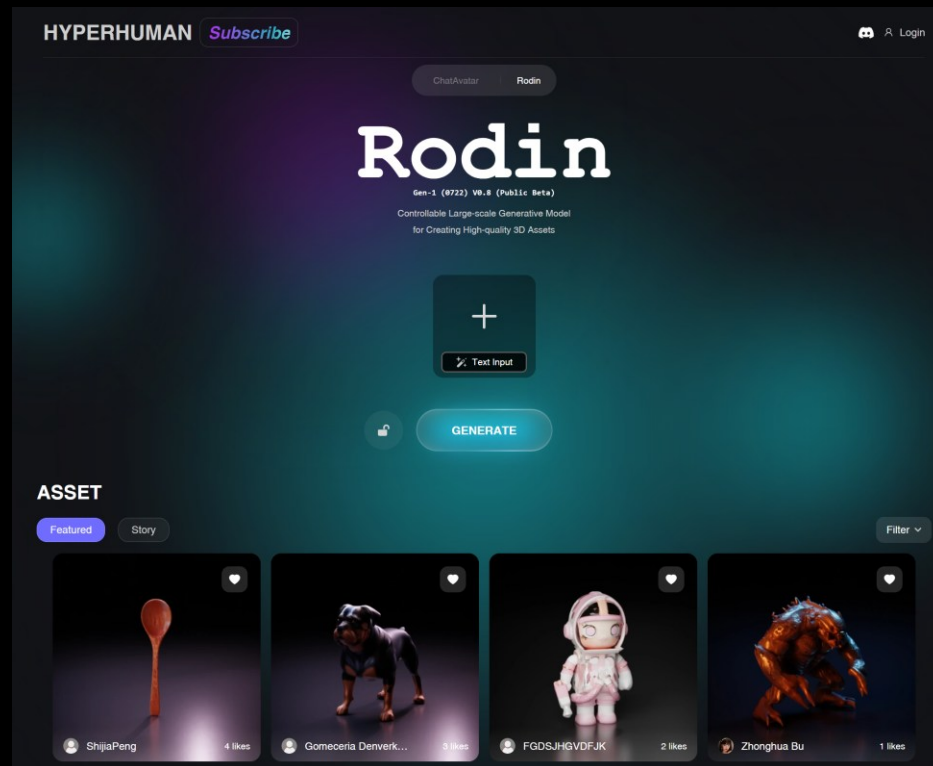
SORA (2024)



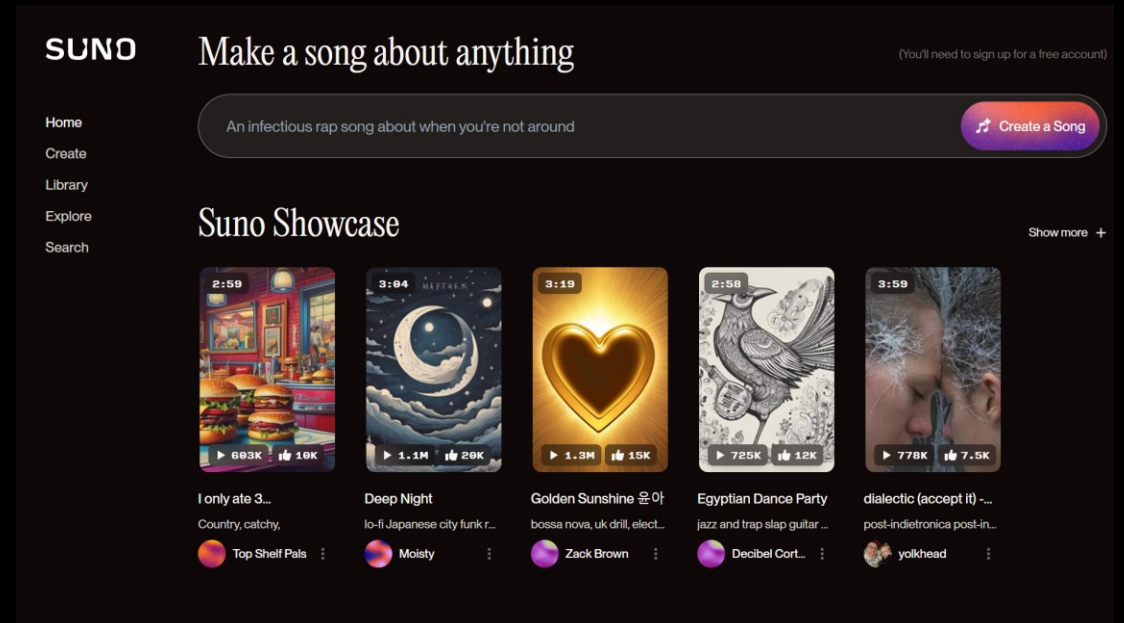
<https://www.youtube.com/watch?v=TuFqvD6PR54>

DEEP LEARNING/NEURAL NETWORK

RODIN (2024)



SUNO(2024)



https://www.youtube.com/watch?v=4w_6HAarpz8

<https://www.youtube.com/watch?v=3JwfFVtSZhU>

DEEP LEARNING/NEURAL NETWORK

- Nano Banana (2025)



DEEP LEARNING/NEURAL NETWORK

- Genie 3 (2025)



CURRENT GEN AI APPLICATIONS

MARKETING

COCA COLA- MASTERPIECE (2023)

Due to the novelty of generative AI,
there are a lot of examples for
advertising to create a fresh visual.



<https://youtu.be/VGa1imApfdg>

ENTERTAINMENT

CYTUS II (2018-PRESENT)

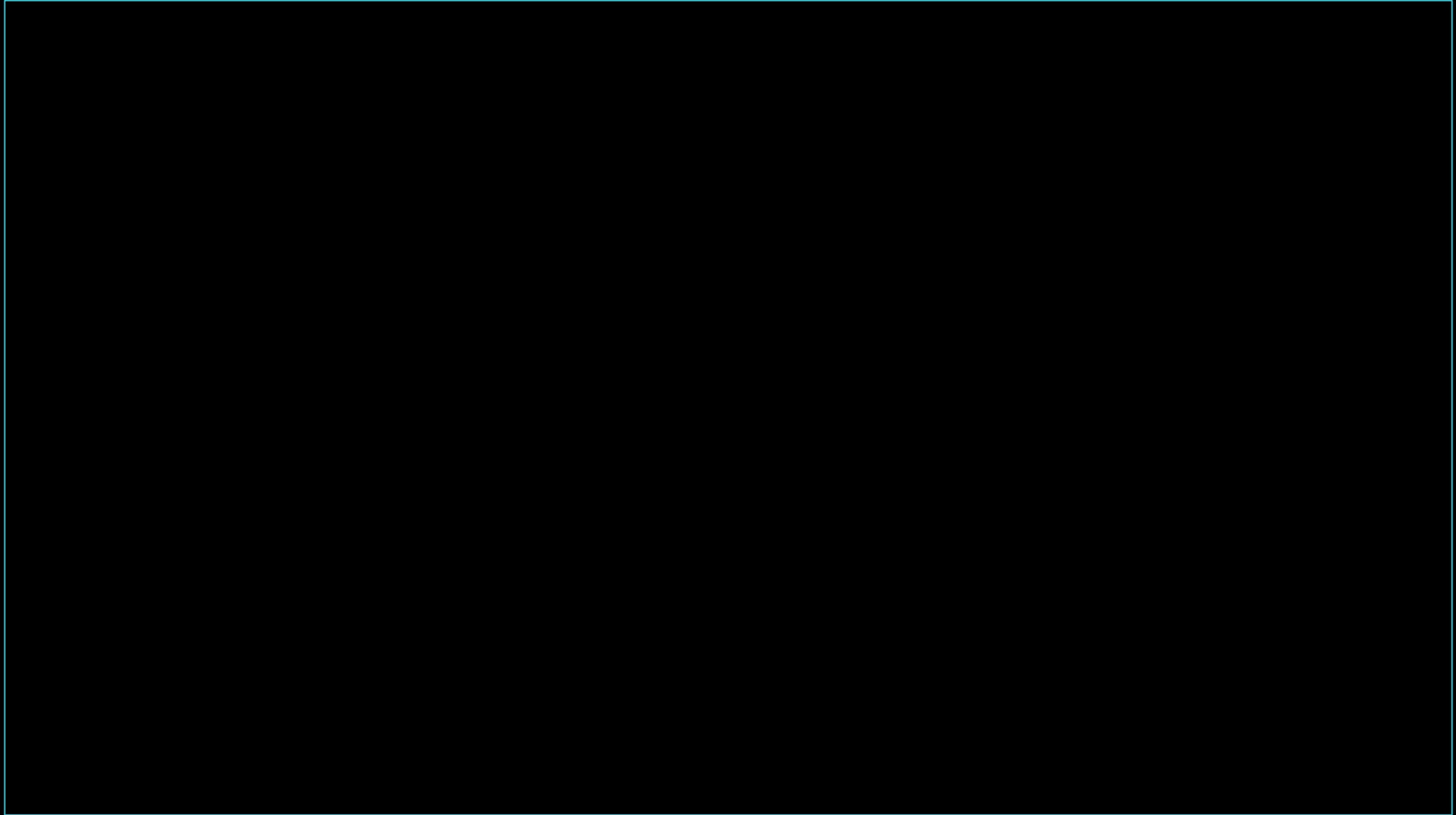
Due to the cost effectiveness of generative AI content, some companies are looking to use it to replace or streamline creating assets in the entertainment industry.



CYTUS II

(2018-PRESENT)

ENTERTAINMENT



ADOBE FIREFLY

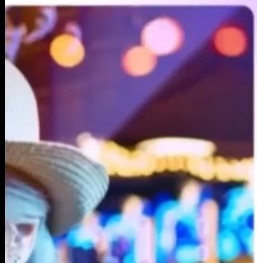
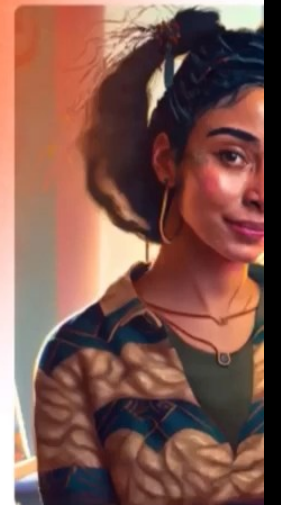
(2023-PRESENT)

TOOLS



 **Adobe Firefly**

firefly.adobe.com



LOTS OF OTHER THINGS

AI ASSISTANTS(AGENTS),
AI ROBOTICS,

...

COURSE SUMMARY

WHAT THE COURSE WILL BE

- AI Progress is going too quickly to cover in a course. New models and new techniques arrive monthly that could make things we talk about quickly obsolete. We aim to cover the core information that can be used to **understand** how these AI tools **may** function.
- AI for design is a broad topic, and we encourage students to approach the domain they are excited with. But for the case of a unified teaching experience, we will focus on the two most popular domains, **text and image**, that are most accessible now.
- Generative AI is currently in a grey area in terms of **safety, ethics, and legality**. We want this course also as a hub for discussion, to talk about what is the best relationship with the technology for a healthy development.
- While we will describe AI Architecture in detail, it will be mostly a high-level description. **We will refrain from complex math but just include the necessary content to help you understand it and talk in a professional way.**

WHAT THE COURSE IS NOT

- Programming knowledge would be tremendously helpful, as it opens you with tools and models that are not already wrapped for ease of use. **We will include some basic Python content to help you program with AI (all self-explanatory).** All tutorials and projects are designed so a minimal knowledge will allow you to work on the tasks.
- Due to the limited resources on hand, we will not be available to provide technical support on domains outside of **LLM and Image Generation**. We expect autonomy of students to solve issues arising from their projects and think it as a learning opportunity to see which area of AI is accessible and which is not.
- **We will not teach how to use design tools like Photoshop, Illustrator, Blender etc.** The focus of this course is to understand how AI tools work under the hood and incorporate into your own workflows.

AI FUNDAMENTALS

- Core AI concepts - Fundamentals
- Architecture of GPT aka Transformers
- Architecture of Image Generation aka Diffusion Models
- Architecture or workflow of advanced techniques:
 - LORA
 - ControlNet, IP-Adapters
 - RAG
 - Vector Database

*Subject to change with course progression

DESIGN APPLICATIONS

- Applying generative AI models to*:
 - Poster Design
 - UX/UI Design (Mid-term mini project)
 - Functional Prototyping
 - Vibe coding

*Subject to change with course progression

LIMITATIONS, SAFETY, ETHICS, AND LEGALITY

- Data Bias
- Fairness
- Hallucination
- Privacy
- Ownership of Copyright in training data and generated output
- Dead Internet
- Labor Replacement
- Current efforts in industry and governments

LAB SCHEDULES

- Lab starts at week 2 with some setup.
- Remember to bring your own laptop.
- We will have a final project and a mini project.

SET LANGUAGE (BETA)



LifeReSTART

IS KEY DEVICE: False

<https://isd3150fp-dev.evnchn.io/>

Life Receipt

 Social Health 100/100

 Mental Health 80/100

 Physical Health 60/100

Every life is extraordinary – no matter what decisions
you've made!

END

BUDGET AND FINAL PROJECT

- Get a group with 2-3 students.
- We might do a group building session to balance technical students with design focus students.
- 500 HKD per student for computation and other service use. *
- Be smart about the budget, don't go crazy initially.
- Reimbursement will be available at the end of course.

* Additional funding may be requestable after consultation

TENTATIVE GRADING SCHEME

- Lab 50% (5 Assignments)
- Mini Project 20% (2 Weeks)
- Final Project 30% (3 Weeks) Proposal 5% and Final Presentation 25%

SYLLABUS

Room: Rm 5510, Lift 25-26

	Session 1 (4:30–6:20 pm, Monday)	Session 2 (3:00–4:50 pm, Friday)	
W1	Introduction & Overview	History of AI & Design	
W2	From Deep Learning to Generative AI	Environment Setup & Python Primary & APIs	A1: text-based image / Generation Test
W3	Chinese New Year Eve	Online Office Hours	Chinese New Year
W4	Design Thinking with AI	Image Generation Lab	A2: Card Design
W5	Brand design with AI	Brand Design Lab	A3: Brand Design
W6	UI & UX & Service Design	UI & UX Design Lab I	Mini project release
W7	Prototype with AI	UI & UX Design Lab II	A4: UI Design
W8	LLM functional prototype (CoT) Lab	RAG & Vector DB	Final project release
W9	RAG & Vector DB Lab	Mini Project Presentation	Mid-term break
W10	Agent and Human-AI Cooperation	Agent Lab	A5: Agent Design
W11	Final Project Proposal	AI Ethics	
W12	AI Safety Lab	Labor Day holiday	
W13	Final Project Presentation	Final Project Presentation	

CONTACTS

- Instructor

Dr. Yuan LIU

Office: Rm 2006, IAS Building (Far away from here)

Email: yuanly@ust.hk

Office Hour: Friday 2:00-4:30 pm. (For other time slots, please contact via emails)

- Teaching Assistants

Mr. Chengwei REN (PhD Student @ ISD)

crenah@connect.ust.hk

Mr. Jiahao LU (PhD Student @ ISD)

jlucn@connect.ust.hk

BACKGROUND SURVEY

- A brief self-introduction (including your full name and nickname, and Y3/4)
- What can AI do for design? Your expectation.
- 2-3 mins talk and 5 mins to prepare~!

BACKGROUND SURVEY – DESIGN

- Sketching/Drawing/Painting
- 2D design (Digital art: Illustrator, photoshop, ...)
- 3D design (e.g. Blender, 3DMax, Maya, Rhino, ...)
- Others: (Video making, Game design, Fashion Design, ...)

BACKGROUND SURVEY – PROGRAMING

- Python
- Web / App Development (HTML)

BACKGROUND SURVEY – MATH

- Vector / Matrix
- Derivatives / Gradient
- Probability Distribution Function (PDF) / Cumulative Distribution Function (CDF) – Gaussian Distribution